

RED CELLS, IRON, AND ERYTHROPOIESIS

Chronic pain in sickle cell disease: a role for lysophosphatidic acid

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KEY POINTS

- HbSS mice have elevated blood levels of LPA, which contributes to sensitization of nociceptive DRG neurons and pain through LPA₁Rs.
- Interrupting LPA-LPA₁R signaling may be a useful approach for managing pain in patients.

Sickle cell disease (SCD) is a hereditary hemoglobinopathy characterized by persistent pain. The mechanisms underlying pain in SCD are poorly understood, and opioids remain the primary treatment, despite their severe side effects. Here, we investigated the contribution of lysophosphatidic acid (LPA), an endogenous pronociceptive lipid mediator, to chronic pain in SCD using humanized transgenic homozygous Berkeley mice that express >99% human sickle hemoglobin (HbSS) and control HbAA mice that express normal human hemoglobin A. Hyperalgesia in HbSS mice was associated with an increase in both plasma level of LPA and expression of LPA receptor 1 (LPA₁R) mRNA in L₁ to L₅ dorsal root ganglion (DRG). Blocking LPA synthesis with BI-2545, or blocking LPA₁R function with small interfering RNA (siRNA) or the LPA₁R antagonist, AM966, reversed mechanical and heat hyperalgesia in HbSS mice. LPA also produced acute mechanical and heat hyperalgesia in HbAA mice, which resulted from the sensitization of C-fiber nociceptors. In HbSS mice, hyperalgesia was associated with sensitization of nociceptive DRG neurons. Nociceptors from hyperalgesic HbSS mice had lower rheobase, more positive

resting membrane potential, and higher frequency of action potential. Although no changes were found in the values of inward currents in nociceptors of HbSS mice compared with HbAA mice, outward-inactivating and noninactivating currents were reduced, indicating the importance of potassium channels to sensitization in SCD. All these parameters were normalized by pretreatment of HbSS mice with LPA₁R siRNA. Our results suggest that LPA signaling may be a promising target for treating pain in SCD.

Introduction

Sickle cell disease (SCD) is the most common inherited monogenic disorder globally.¹ SCD results from a point mutation in the β -globin gene, leading to the replacement of normal hemoglobin A (HbA) with hemoglobin S (HbS).² HbS homozygotes (HbSS) suffer from the most severe form of SCD. HbS polymerizes during deoxygenation, causing deformation of erythrocytes ("sickling"), loss of elasticity, and their adherence to vascular endothelium. This produces vaso-occlusion and severe acute pain, referred to as vaso-occlusive crisis (VOC).³⁻⁵ Pain during VOC often requires hospitalization and treatment with opioids.^{3,6} The cascade of recurrent acute VOCs with age leads to persistent inflammation, organ damage, and chronic pain⁷⁻¹⁰; and chronic pain can exist without the presence of VOCs.¹¹ Moreover, chronic pain can persist after successful gene therapy,¹² suggesting long-term irreversible changes in pain signaling. Although adolescents and young adults with SCD experience predominantly acute pain,

adult patients exhibit both acute and chronic pain¹³ that complicates pain management.

Berkeley humanized transgenic (BERK) mice, referred to as sickle (HbSS) mice, are widely used to study the mechanisms of SCD and test new treatments.¹⁴ As in patients with SCD, HbSS mice exhibit hemolytic anemia, inflammation, VOC, organ damage,^{8,9,13,15,16} and chronic pain, as indicated by the presence of chronic mechanical and thermal hyperalgesia.¹⁷⁻²¹ Hyperalgesia is mediated by the sensitization of nociceptive dorsal root ganglion (DRG) and dorsal horn neurons,^{19,22-24} arising after exposure to inflammatory mediators and tissue damage products,²⁵ including lysophosphatidic acid (LPA).

Extracellular LPA is a powerful pronociceptive mediator²⁶ contributing to inflammatory and neuropathic pain in animal models.^{27,28} In humans, elevated LPA is associated with chronic neuropathic pain²⁹ and its level in cerebrospinal fluid correlates with the intensity of neuropathic pain.³⁰

Intracutaneous injection of LPA in humans produced burning pain, hyperalgesia, and sensitization of C-fiber nociceptors,³¹ and intrathecal (IT) injection of LPA produced acute nociceptive behaviors in rodents.^{32,33} The bulk of extracellular LPA is mainly produced by the ectoenzyme autotaxin (ATX) from membrane lysophospholipids as precursors.^{34,35} Increased ATX was associated with inflammatory and neuropathic pain in patients^{36,37} and hyperalgesia in a mouse model of cancer pain.³⁸ ATX inhibition reduces LPA production, leading to a decrease in LPA levels and subsequent pain relief.^{37,38}

Among the 6 G protein-coupled receptors involved in LPA signaling, LPA receptor 1 (LPA₁R) is the main subtype expressed in DRG neurons and involved in pain.^{32,39} Inhibition of LPA₁R reduced both inflammatory and neuropathic pain.^{27,40} By modulating certain types of ion channels,⁴¹ including inhibition of voltage-dependent and 2-pore domain potassium (K⁺) channels,⁴² LPA-LPA₁R signaling increases the excitability of small nociceptive DRG neurons, thereby causing pain.^{43,44} Both channels are widely expressed in small DRG neurons⁴⁵⁻⁴⁹ and regulate resting membrane potential (RMP), action potential (AP) frequency, and membrane repolarization after AP generation.⁵⁰

Here, we examined the contribution of LPA-LPA₁R signaling to chronic pain in mice with SCD. Our results show that serum levels of LPA are elevated in HbSS mice, and LPA contributes to hyperalgesia by sensitizing small DRG neurons via LPA₁R. Thus, targeting LPA-LPA₁R signaling may offer a new approach for treating chronic pain in SCD.

Materials and methods

Animals

In all experiments, an equal number of male and female transgenic homozygous HbSS-BERK and HbAA-BERK (aged 6-9 months) mice were used. Mice were maintained on a 12-hour light-dark schedule and had food and water ad libitum. All procedures were approved by the University of Minnesota Institutional Animal Care and Use Committee. At the end of experiments, mice were euthanized by inhalation of CO₂. All the results presented in the article contain data obtained equally from male and female mice, unless indicated otherwise.

Drugs

For all experiments, oleoyl LPA (18:1-LPA) was purchased from Cayman Chemical in the form of a 22.9mM solution in ethanol. AM966 (2-[4-[4-[(1R)-1-(2-chlorophenyl)ethoxy]carbonylamino]-3-methyl-1,2-oxazol-5-yl]phenyl]phenyl]acetic acid), an LPA₁R antagonist, was purchased from Cayman Chemical. A stock solution (10 mg/500 µL) was prepared in dimethyl sulfoxide and diluted in saline to achieve the final dose. The ATX inhibitor, BI-2545, was purchased from MedChemExpress and diluted in accordance with their recommendations. All other chemicals were purchased from Sigma-Aldrich.

Behavioral measures of hyperalgesia

Mechanical hyperalgesia was defined as a decrease in paw withdrawal threshold determined for each hind paw using calibrated von Frey (VF) monofilaments (Stoelting) as previously described,⁵¹ or as an increase in the frequency of paw

withdrawal evoked by a VF monofilament with a bending force of 3.9 mN.¹⁹ This monofilament was applied 10 times, each for 2 seconds, to random locations on the plantar surface of each hind paw. The number of withdrawal responses evoked by the monofilament was expressed as a percentage. Heat hyperalgesia was defined as a decrease in paw withdrawal latency (PWL) in response to a radiant heat stimulus applied to the plantar surface of each hind paw as we have previously described.¹⁹ Mice were placed on an elevated glass platform, covered with glass containers, and habituated for 15 minutes. Radiant heat was applied to the plantar surface of each hind paw and the mean PWL was calculated from 3 trials. The intensity of the heat source was adjusted so that mice withdrew their hind paws at ~9 seconds during baseline testing. A cutoff time of 20 seconds was chosen to avoid tissue damage.

Measures of spontaneous nociceptive behaviors

Spontaneous nociceptive behaviors were assessed by the parameters of ear position (the angle of ear inclination relative to the head), eye eccentricity (orbital tightening), and body eccentricity (decreased length and increased curvature) values calculated according to the formula: $e = [\sqrt{(a^2 - b^2)}]/a$, in which "a" is the major axis and "b" is the minor axis of ellipses defined for eye and body shape.^{52,53} Mice were video-recorded for 5 minutes. Only images that provided full access to the eyes and body position were selected. Scoring was done by 3 blinded observers. For all behavioral studies, experimenters were blind to mice genotypes and treatments.

Serum isolation

Blood (0.5 mL) was collected and kept for 30 minutes for clotting. Serum was isolated by centrifugation for 10 minutes at 2000g at 4°C. The supernatant was used for measurement of LPA. Samples were stored at -80°C until processing.

Measurement of endogenous LPA

The total serum level of LPA was quantified as the sum of the 6 identified LPA species using ultraperformance liquid chromatography (UPLC)-tandem mass spectrometry, as we have described in detail elsewhere.³⁸ Briefly, after extraction, LPA species were resolved on an ACQUITY UPLC HSS T3 column (1.8 µm; pore diameter, 100 Å; 2.1 × 150 mm, at 55°C) with an ACQUITY UPLC Waters system using a gradient of 2mM of ammonia formate in water as solvent A and 2mM of ammonia formate in methanol as solvent B at a flow rate of 0.3 mL/min. A Waters Xevo TQ-S triple-quadrupole mass spectrometer operated in a negative ionization mode was used for LPA detection. The mass transitions were: 16:0-LPA, 409.2/152.9; 18:0-LPA, 437.2/152.9; 18:1-LPA, 435.2/152.9; 18:2-LPA, 433.2/152.9; 18:2-LPA-d5, 438.3/157.9; 20:4-LPA, 457.2/152.9; and 22:6-LPA, 481.2/152.9. LPA species were quantified against d5-18:2-LPA internal standard using standard curves and expressed in micrograms per milliliter of serum. The effect of BI-2545 on serum LPA levels was determined using an enzyme-linked immunosorbent assay LPA kit (MyBioSource) in accordance with the manufacturer's protocol.

RT-qPCR

LPA₁R, LPA₃R, and LPA₅R mRNAs were measured in the L₁ to L₅ DRG and lumbar spinal cord enlargement from HbAA and HbSS mice using quantitative real-time polymerase chain

reaction (RT-qPCR). After RNA isolation using Trizol reagent (Thermo Fisher Scientific), samples were incubated with TURBO DNase (Ambion). RNA was purified with the Zymo Research RNA Clean and Concentrator kit (Thomas Scientific) and quantified by spectrophotometric analysis with a NanoDrop 2000 (Thermo Scientific). Complementary DNA was synthesized using 1 µg of mRNA sample with a Reverse Transcriptase Protoscript II kit (New England BioLabs). RT-qPCR assay was performed using the Maxima SYBR Green/ROX RT-qPCR Master Mix (Thermo Scientific). A QuantStudio 3 (Applied Biosystems) was used for quantification of the PCR reaction. All kits were used according to the manufacturers' protocol. Detected transcript levels were normalized to the β -Actin housekeeping gene. Primer pair sequences were as follows: LPA₁R (IDT), F' CTATGTTCCGCCAGAGGACTATG and R' GCAATAACAAGACCAATCCCG; LPA₃R (IDT), F' ACCAACGTCTTATCTCCACAC and R' CAGTTCAGGCCGTC-CAG; LPA₅R (IDT), F' TGGAGGTGAAAGTCATGCTC and R' GTATCTCGAYAGTCAGGGCAC.

Delivery of LPA₁R siRNA

Scrambled or LPA₁R small interfering RNA (siRNA; Thermo Fisher) were administered IT (1.5 µg siRNA in I-Fect reagent [Neuromics Inc] in a weight-to-volume ratio of 1:4) on 3 consecutive days according to the manufacturer's protocol and previous publications.⁵⁴⁻⁵⁶ The effectiveness of siRNA in reducing LPA₁R expression treatment was confirmed by RT-qPCR.

Electrophysiology

Electrophysiological recordings were made from single nociceptive C-fibers from the tibial nerve in HbAA mice, as described previously.¹⁹ Mice were anesthetized with isoflurane, and the tibial nerve was isolated. Nerve fascicles were teased apart and placed on a stainless-steel recording electrode. Extracellular recordings were obtained only from single fibers discriminated according to amplitude and shape of their AP. The receptive field (RF) of each fiber was identified using a calibrated suprathreshold VF monofilament (149 mN force). Nociceptors were identified by their responses to different types of mechanical stimuli including light touch (stroking with a cotton-tipped applicator) and mild pinch. Fibers responding to pinch, but not light touch, were studied. Conduction velocity was determined by electrically stimulating the RF and measuring the conduction latency and the distance between the RF and the recording electrode. Fibers with a conduction velocity of <1.3 m/s were considered C-fibers.

Spontaneous activity (SA; 1 minute) and responses evoked by controlled noxious mechanical and heat stimuli applied to the RF were obtained before and at 60 minutes after IV injection of LPA (1.4 µg/kg), or vehicle (4.2% of ethanol in saline). Calibrated VF monofilaments were used to deliver mechanical stimuli. Mechanical response threshold (millinewtons) was defined as the minimum force that evoked a response in at least 50% of the trials. Responses evoked by the 149 mN VF monofilament applied to the RF for 5 seconds were expressed as the mean of 3 trials, each separated by 60 seconds. To further characterize nociceptors functionally, a Peltier device (10 mm² probe) was used to deliver computer-controlled heat (50°C for 5 seconds) to the RF from a base temperature of 32°C and a ramp rate of 2°C/s.

Patch-clamp electrophysiology

Passive and active electrical membrane properties of acutely isolated small L₁ to L₅ DRG neurons obtained from hyperalgesic HbSS mice and control (nonhyperalgesic) HbAA mice were studied by patch-clamp recording. To dissociate neurons, DRGs were incubated with dispase II (5 mg/mL) and collagenase I (2 mg/mL) for 45 minutes at 37°C and plated on glass coverslips coated with laminin, as we have described elsewhere.⁵⁷ Passive membrane properties included membrane capacitance and input resistance, and active membrane properties included inward current, outward-inactivating, and noninactivating currents; RMP; rheobase; and AP threshold, duration, and frequency. Studies were conducted using a bath solution containing: NaCl (145mM), KCl (5mM), MgCl₂ (2mM), CaCl₂ (2mM), glucose (5mM), and HEPES (*N*-2-hydroxyethylpiperazine-*N'*-2-ethanesulfonic acid; 10mM), pH 7.3 and 330 mOsm. The internal electrode solution contained K-gluconate (105mM), KCl (30mM), CaCl₂ (0.5mM), MgCl₂ (2.4mM), KCl (5mM), EGTA (5mM), HEPES (10mM), Na₂-ATP (2mM), Na₂-GTP (0.33mM), pH 7.2 and 335 mOsm.⁵⁸ SA was recorded using a gap-free protocol and resistance of the recording electrode was ~5 MΩ. The junction potential was 13.7 mV and was adjusted during data analysis. The series resistance of each recording was <30 MΩ. Recording signals were amplified with Multiclamp 700B amplifier (Axon Instruments), filtered at 2 kHz, and sampled at 5 kHz using pClamp 10 software (Axon Instruments). For voltage-clamp recordings, the holding potential was -73.7 mV, including the correction of the junction potential. Voltage stimuli from -93.7 to +46.3 mV, with a duration of 500 milliseconds, were delivered in increments of 10 mV. For recordings in current-clamp mode, step current pulses were injected into cells through the patch-clamp electrode. Currents from -20 pA to 2000 pA were delivered in increments of 10 pA per step, each for a duration of 1 second.

Statistical analysis

Normally distributed data are presented as mean ± standard error of the mean. These data were analyzed using Student *t* test or 1- or 2-way analysis of variance (ANOVA) with repeated measures ANOVA as applicable. Post hoc analyses were performed using Bonferroni post hoc tests to determine differences between specific groups. Data that did not pass a normality test are presented as median and were analyzed using Mann-Whitney rank-sum test or Kruskal-Wallis 1-way ANOVA on ranks followed by Dunn or Tukey tests to determine differences between groups. The Pearson correlation coefficient determined the relationship between LPA₁R mRNA levels and hyperalgesia. The χ^2 test was used to determine differences in the percentage of neurons exhibiting SA between groups. Statistical analyses were performed using SigmaPlot 14.6 software. *P* values <.05 were considered significant.

Results

Hyperalgesia in HbSS mice was accompanied by an increase in serum LPA

Consistent with previous reports,^{19,22,24,59,60} HbSS mice exhibited pronounced chronic mechanical and heat hyperalgesia (Figure 1A-B) without signs of spontaneous nociceptive behavior as determined by changes in body eccentricity

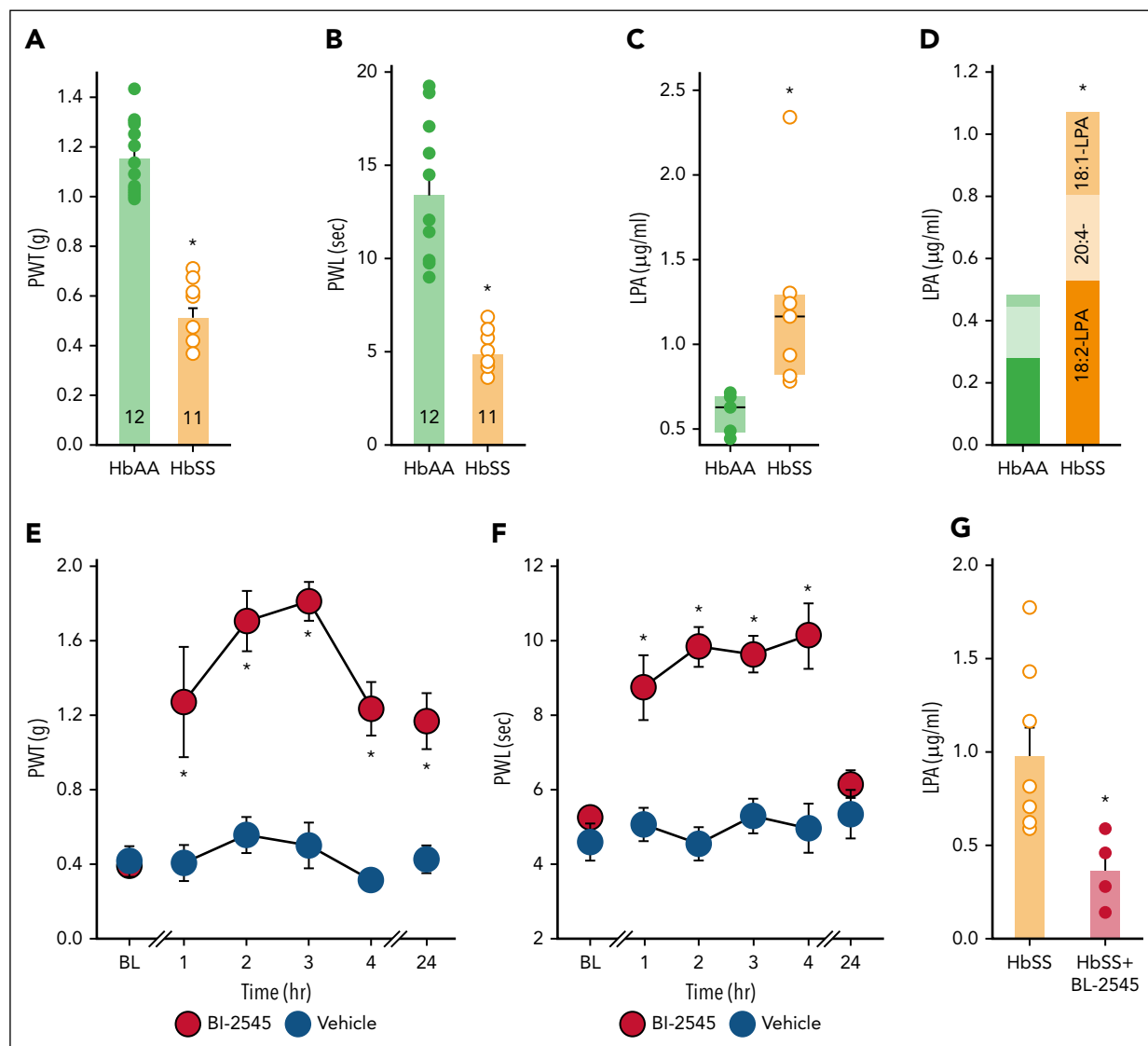


Figure 1. LPA contributes to hyperalgesia in HbSS mice. HbSS mice had lower PWT and PWL values than control HbAA mice, indicating mechanical (A) and thermal (B) hyperalgesia, respectively. *Different from HbAA mice at $P < .001$; Student *t* test. Numbers inside bars indicate group size. (C) Hyperalgesia was accompanied by an increase in the level of LPA in serum measured using UPLC–tandem mass spectrometry and presented as the sum of the 6 identified LPA species. *Different from HbAA mice at $P = .001$; Mann-Whitney rank-sum test; $n = 6$ to 7 mice per group. Only 18:2-, 20:4-, and 18:1-LPA were elevated in HbSS mice (D). *Different from the corresponding LPA species in HbAA mice ($P = .012$, $P = .042$, Student *t* test; and $P = .001$, Mann-Whitney test, respectively). IP administration of the ATX inhibitor, BI-2545 (10 mg/kg), blocked mechanical (E) and heat (F) hyperalgesia in HbSS mice. A decrease in mechanical hyperalgesia was observed 1 hour after injection and persisted for 24 hours ($F[5,45] = 5.94$; $P < .001$; $n = 5$ –6 mice per group; 2-way repeated measures [RM] ANOVA). *Different from baseline within BI-2545 and from vehicle at $P < .001$ (2-way RM ANOVA with Bonferroni test). A decrease in heat hyperalgesia was observed 1 hour after injection of BI-2545 and persisted for 4 hours. Hyperalgesia was restored 24 hours after the injection ($F[5,70] = 5.34$; $P < .001$; $n = 5$ –12 mice per group; 2-way RM ANOVA). The vehicle, dimethyl sulfoxide (DMSO), PEG400, Tween-80, and saline (10:40:5:45) did not alter mechanical or heat hyperalgesia ($P = 1.0$). *Different from baseline within BI-2545 and vehicle at $P < .001$ (2-way RM ANOVA with Bonferroni test). The antihyperalgesic effect of BI-2545 was accompanied by a decrease in serum LPA in HbSS mice measured by an enzyme-linked immunosorbent assay LPA kit 3 hours after administration (G). *Different from HbSS at $P = .007$, Student *t* test; $n = 4$ to 8 mice per group. BL, baseline; PWT, paw withdrawal threshold.

($P = .369$, Mann-Whitney test), eye eccentricity ($P = .741$, Mann-Whitney test), and ear angle ($P = .249$, Student *t* test, $n = 6$ mice per group). Hyperalgesia was accompanied by an increase in the level of LPA in serum, presented as the sum of individual LPA species (Figure 1C). Of the 6 identified LPA species, 18:2-, 20:4-, and 18:1-LPA were elevated in HbSS mice (Figure 1D). All 3 species are considered strong LPA₁R agonists, with 18:1-LPA showing the highest affinity.⁶¹ The levels of 16:0-, 18:0-, and 22:6-LPA species in HbSS mice were similar to those in HbAA mice ($P = .324$, $P = .371$, and $P = .628$, Student *t* test; $n = 6$ –7 mice per group).

The contribution of LPA to chronic hyperalgesia in HbSS mice was confirmed by inhibiting its synthesis. Intraperitoneal (IP) administration of BI-2545 (10 mg/kg, IP), an inhibitor of ATX, the main enzyme that synthesizes the extracellular pool of LPA, blocked mechanical and heat hyperalgesia in HbSS mice as compared with vehicle (Figure 1E–F). Consistent with previous pharmacokinetic studies showing a rapid reduction in plasma LPA that persisted for at least 8 hours,⁶² a single dose of BI-2545 decreased the total serum level of LPA in HbSS mice at 3 hours after administration (Figure 1G) and was previously effective in a mouse model of bone cancer pain.³⁸ BI-2545 did

not alter withdrawal responses in HbAA mice ($F[6,24] = 0.765$, 1-way repeated measures ANOVA; $P = .605$; $n = 5$ mice per group).

LPA-induced hyperalgesia is due to sensitization of C-fibers

The acute pronociceptive effect of circulating LPA was determined in control HbAA mice. Intravenous administration of LPA (1.4 mg/kg, IV, the dose equivalent to the maximum level of LPA determined in the serum of hyperalgesic HbSS mice) produced acute mechanical and heat hyperalgesia in HbAA mice that persisted for 3 hours after administration (Figure 2A-B), and this was accompanied by sensitization of C-fiber nociceptors. LPA (1.4 mg/kg, IV), but not vehicle, evoked SA of C-fibers (Figure 2C-D), decreased their mechanical response thresholds (Figure 2E), increased responses to mechanical stimuli (Figure 2F-G), decreased heat response thresholds (Figure 2H) and increased responses to heat (Figure 2I-J). These data show that hyperalgesia produced by LPA was due to sensitization of C-fiber nociceptors.

LPA-LPA₁R signaling in DRGs contributes to hyperalgesia in HbSS mice

Although both DRG neurons and satellite cells express LPA₁, LPA₃, and LPA₅ receptors,^{32,38,63-66} only LPA₁R mRNA was upregulated in L₁ to L₅ DRGs from HbSS mice as compared with HbAA mice (Figure 3A). No changes were observed in mRNA for LPA₃R and LPA₅R subtypes ($P = .789$ and $P = .975$, Student *t* test; $n = 7$ mice per group). Furthermore, no changes in the expression of these 3 LPARs were observed in the spinal cord of HbSS mice compared with HbAA mice (LPA₁R: $P = .366$, LPA₃R: $P = .947$ and LPA₅R: $P = .960$, Student *t* test; $n = 4$ mice per group). IT administration of LPA₁R siRNA, but not the scrambled siRNA, reduced the level of LPA₁R mRNA in L₁ to L₅ DRGs (Figure 3B) but not in the spinal cord ($F[2,7] = 0.591$, 1-way ANOVA; $P = .579$; $n = 3-4$ mice per group), probably due to differences in siRNA uptake and the presence of a blood-brain barrier. A decrease in LPA₁R mRNA in DRGs was accompanied by a decrease in mechanical and heat hyperalgesia in HbSS mice (Figure 3C-D), with a strong positive correlation between LPA₁R mRNA levels and changes in paw withdrawal threshold and PWL (Pearson coefficient = 0.94; $P = .006$ and $.005$, respectively). Furthermore, a single IP injection of AM966 (30 mg/kg), an LPA₁R antagonist,^{67,68} but not the vehicle, also reduced chronic mechanical and heat hyperalgesia in HbSS mice (Figure 3E-F). Collectively, these data show that elevated circulating LPA contributes to hyperalgesia in HbSS mice through LPA₁R signaling in DRGs.

LPA-LPA₁R signaling promotes nociceptor sensitization in HbSS mice by decreasing K⁺ currents

Acutely dissociated DRG neurons were used to investigate the contribution of LPA₁R signaling to nociceptor sensitization in HbSS mice and underlying biophysical mechanisms. The integral of the membrane capacitive current recorded during a 10 mV pulse without active current was used to calculate the capacitance. Only small neurons (membrane capacitance of <40 pF) were used. Nociceptive neurons were identified by the presence of a shoulder in the repolarization phase of their AP (Figure 4A), established using the first derivative of the AP

waveform.^{57,69} Nociceptive neurons from hyperalgesic HbSS mice had lower rheobase, increased RMP, and a higher frequency of APs than nociceptors from HbAA mice (Figure 4B-D). There was no difference in inward currents (Figure 4E), whereas outward-inactivating and noninactivating currents, which are mediated by various K⁺ channels and are responsible for the repolarization of neurons, were reduced in nociceptors from HbSS mice (Figure 4F-G). Thirty percent of nociceptive DRG neurons in HbSS mice exhibited SA, which was not observed in any nociceptors in HbAA mice ($P < .001$, χ^2 test; $n = 10$ neurons per group). Importantly, these electrophysiological parameters, as well as SA ($P < .001$, χ^2 test), were normalized in nociceptive neurons from HbSS mice treated with LPA₁R siRNA (Figure 4B-D,F-G), indicating the contribution of LPA₁R to these electrophysiological changes. No differences were found in the input resistance, AP amplitude or AP duration ($P = .23$, $P = .84$, $P = .35$, respectively; Student *t* test). The absence of changes in inward currents and AP threshold ($P = .67$, Student *t* test) indicated that sodium channels are not involved in sensitization of DRG neurons in HbSS mice. These data suggest that LPA₁R-mediated sensitization of small nociceptive neurons in HbSS mice is primarily due to a slowing of the repolarization process resulting from a decrease in K⁺ currents.

Discussion

In this study, we showed that elevated serum LPA, an important source of extracellular LPA, contributes to chronic mechanical and thermal hyperalgesia in a humanized mouse model of SCD due to nociceptor sensitization. Inhibition of ATX, the main enzyme producing extracellular LPA,³⁵ reduced serum LPA levels and blocked mechanical and thermal hypersensitivity. Furthermore, IV administration of LPA sensitized C-fiber nociceptors and caused hyperalgesia in control HbAA mice, confirming its pronociceptive effect. It is generally accepted that production of extracellular LPA is triggered by tissue damage and/or chronic inflammation, with ATX and phospholipid precursors playing a crucial role in this process.⁷⁰ An increase in the level of serum LPA may be associated with platelet activation,⁷¹ which occurs in patients with SCD at steady state and especially during VOCs.^{72,73} Patients with SCD exhibited elevated levels of LPA precursors in red blood cells (RBCs) and bone marrow, which correlated with VOC severity.^{74,75} Elevated LPA in serum may be a common marker of chronic pain because this positively correlated with pain intensity in patients with fibromyalgia and osteoarthritis,³⁰ and in a model of bone cancer pain.³⁸ Although our data strongly suggest the contribution of circulating LPA to chronic hyperalgesia in HbSS mice, local production of extracellular LPA in DRG neurons cannot be excluded, because they express ATX.²⁷ Interestingly, our studies did not reveal visual signs of spontaneous nocifensive behavior in hyperalgesic HbSS mice at steady state assessed by ear position, and eye and body eccentricity, although ongoing spontaneous pain was previously reported in HbSS mice using a conditioned place preference test.⁷⁶ Because visual expression of pain was previously reported in HbSS mice during acute cold-induced painful VOC,^{52,53} it is possible that chronic pain, which is characterized by neural plasticity, reduces the visual manifestation of pain.⁷⁷

All LPA species elevated in the serum of HbSS mice (18:2-, 20:4-, and 18:1-LPA) are considered strong LPA₁R agonists,

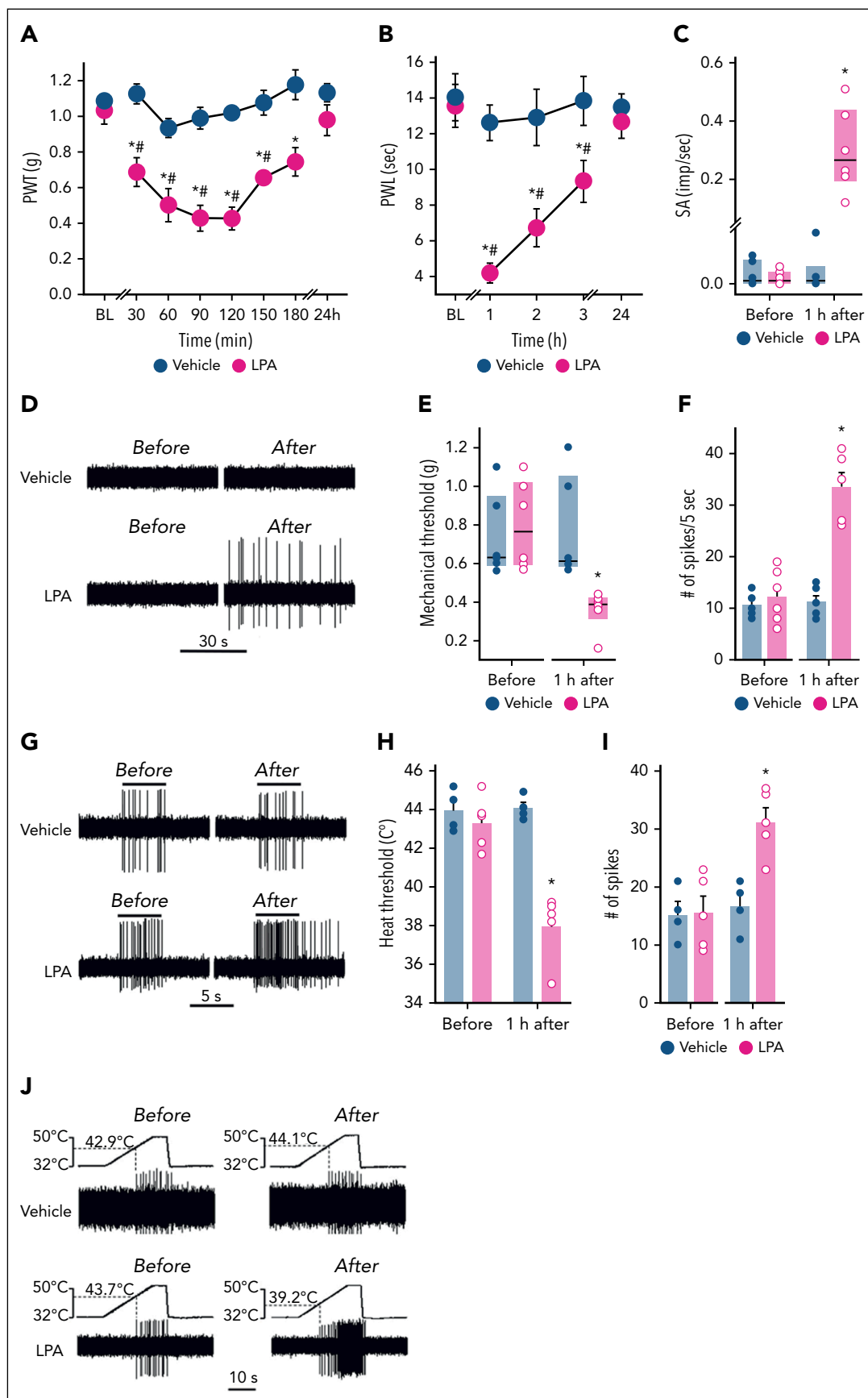


Figure 2.

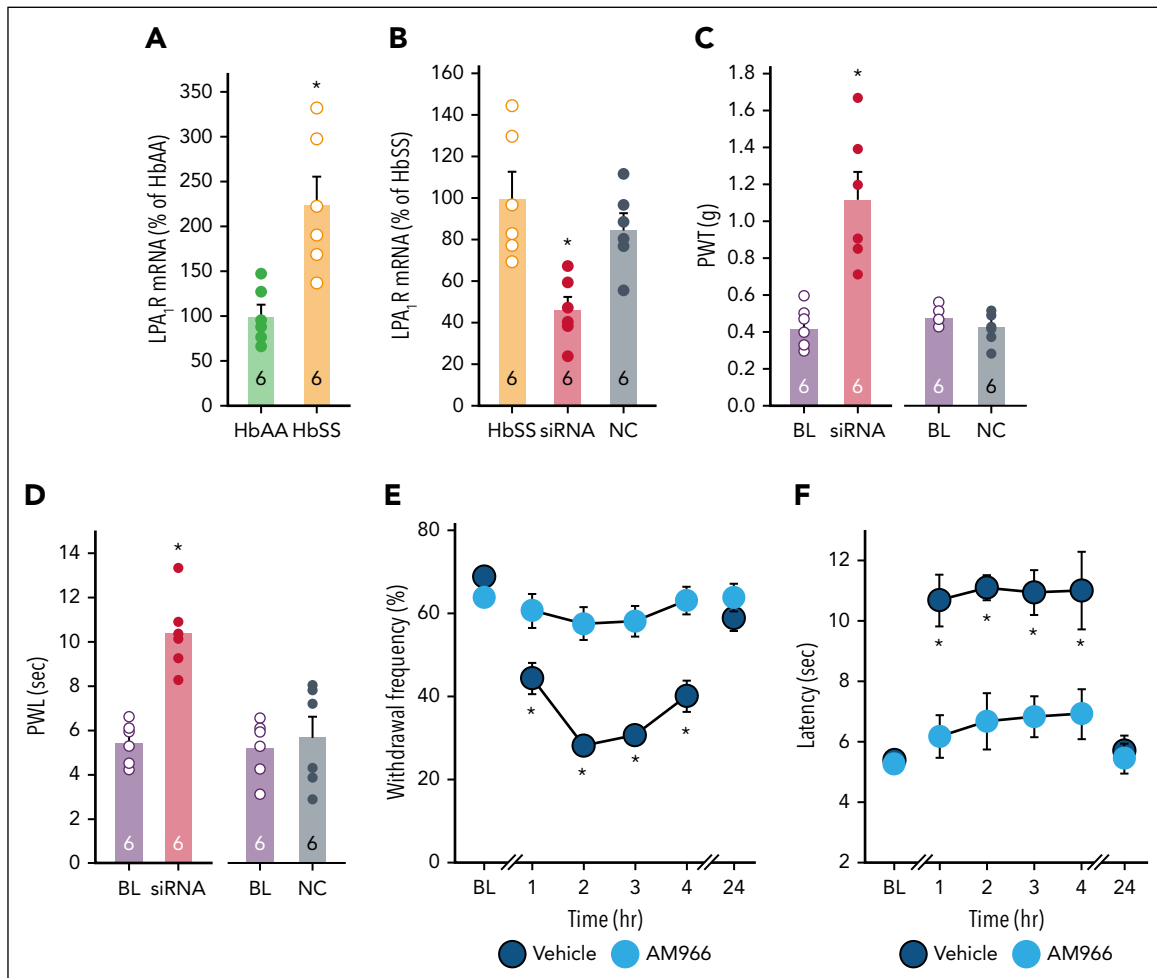


Figure 3. LPA₁R signaling contributes to hyperalgesia in HbSS mice. (A) LPA₁R mRNA expression was higher in L₁ to L₅ DRGs from hyperalgesic HbSS mice than HbAA mice. *Different from HbAA mice at $P = .008$, Student t test. Numbers inside bars indicate group size. (B) Three consecutive daily IT injections of LPA₁R siRNA (1.5 μ g of siRNA in I-Fect reagent in a weight-to-volume ratio of 1:4) significantly reduced LPA₁R mRNA expression compared with untreated HbSS mice. *Different from other groups ($F[2,15] = 9.09$; $P = .003$, 1-way ANOVA with Bonferroni test). Scrambled siRNA (NC) had no effect ($P = .79$). (C) LPA₁R siRNA suppressed mechanical hyperalgesia in HbSS mice, as evidenced by an increase in PWT values. *Different from other groups ($F[1,10] = 19.19$; $P = .001$, 2-way RM ANOVA with Bonferroni test). The scrambled siRNA (NC) had no effect ($P = 0.67$). (D) Administration of LPA₁R siRNA also decreased heat hyperalgesia in HbSS mice as indicated by an increase in PWT to heat. *Different from other groups ($F[1,10] = 13.42$; $P = .004$, 2-way RM ANOVA with Bonferroni test). The NC had no effect ($P = 0.58$). (E) Systemic administration of the LPA₁R antagonist, AM966 (30 mg/kg, IP), decreased mechanical hyperalgesia in HbSS mice, defined as a decrease in paw withdrawal frequency ($F[5,70] = 10.70$; $P < .001$, 2-way RM ANOVA; $n = 8$ mice per group). Vehicle was DMSO in saline (22.5:75.5, v:v) and had no effect. *Different from baseline within AM966 and from vehicle at $P < .001$ (2-way RM ANOVA with Bonferroni test). (F) AM966 also decreased chronic heat hyperalgesia in HbSS mice ($F[5,55] = 7.25$; $P < .001$, 2-way RM ANOVA; $n = 6-5$ mice per group). *Different from baseline within AM966 and from vehicle at $P < .001$ (2-way RM ANOVA with Bonferroni test). BL, baseline; NC, negative control; PWT, paw withdrawal threshold.

with 18:1-LPA showing the highest affinity.⁶¹ High agonist potency of 18:1-LPA on LPA₁R was reported in models of inflammatory⁷⁸ and neuropathic³³ pain. Hyperalgesia in HbSS mice was accompanied by increased LPA₁R expression in DRGs but not in the spinal cord. Knockdown of LPA₁R by siRNA or its inhibition with AM966, eliminated mechanical and heat

hyperalgesia in HbSS mice, which opens new possibilities in managing pain in patients with SCD. Although our experiments suggest that the knockdown of LPA₁R mRNA in DRGs after IT administration of the siRNA was due to uptake by the central terminals of DRG neurons and point to the periphery as the main site of LPA action in inflammatory pain, the possible

Figure 2. LPA produced hyperalgesia that is due to sensitization of C-fiber nociceptors. IV administration of LPA (1.4 mg/kg, IV) produced acute mechanical hyperalgesia in HbAA mice (A), defined as a decrease in PWT ($F[1,56] = 91.58$; $P < .001$; $n = 6$ mice per group; 2-way RM ANOVA). *Different from baseline at $P < .05$, and #different from vehicle at $P < .01$ with Bonferroni test. Vehicle (DMSO, Tween-80, and saline, 20:2:78, v:v:v) had no effect ($P = 1$). LPA, but not vehicle, produced heat hyperalgesia in HbAA mice (B), defined as a decrease in PWT ($F[1,36] = 20.24$; $P = .001$; $n = 6$ mice/treatment; 2-way RM ANOVA). *Different from baseline at $P < .05$, and #different from vehicle at $P < .01$ with Bonferroni test. LPA-induced hyperalgesia was accompanied by an increase in SA of C-fibers (C) ($P = .003$), decreased mechanical thresholds (E) ($P = .004$, 1-way ANOVA with the Dunn test, $n = 6$ fibers per group), and an increase in their responses (number of APs) to mechanical stimuli (F) (147 mN for 5 seconds; $F[3,20] = 39.06$; $P < .001$, 1-way ANOVA with Bonferroni test; $n = 6$ mice per group). *Different from other groups. LPA, but not vehicle, decreased heat thresholds (F[3,14] = 23.50; $P < .001$, 1-way ANOVA with Bonferroni test) (H) and increased C-fiber responses to heat stimuli (ramp from 32 to 50°C; $F[3,14] = 9.81$; $P < .001$, 1-way ANOVA with Bonferroni test, $n = 4-6$ fibers per group) (I). Representative examples of SA (D), and responses to mechanical (G) and heat (J) stimuli for single C-fibers before and after vehicle and LPA are provided. Time of mechanical and heat stimuli are indicated by the solid lines. The heat response threshold is provided and defined by the dotted line (temperature that evokes the first AP). BL, baseline; PWT, paw withdrawal threshold. For all panels, blue refers to vehicle-treated groups and pink refers to LPA-treated groups.

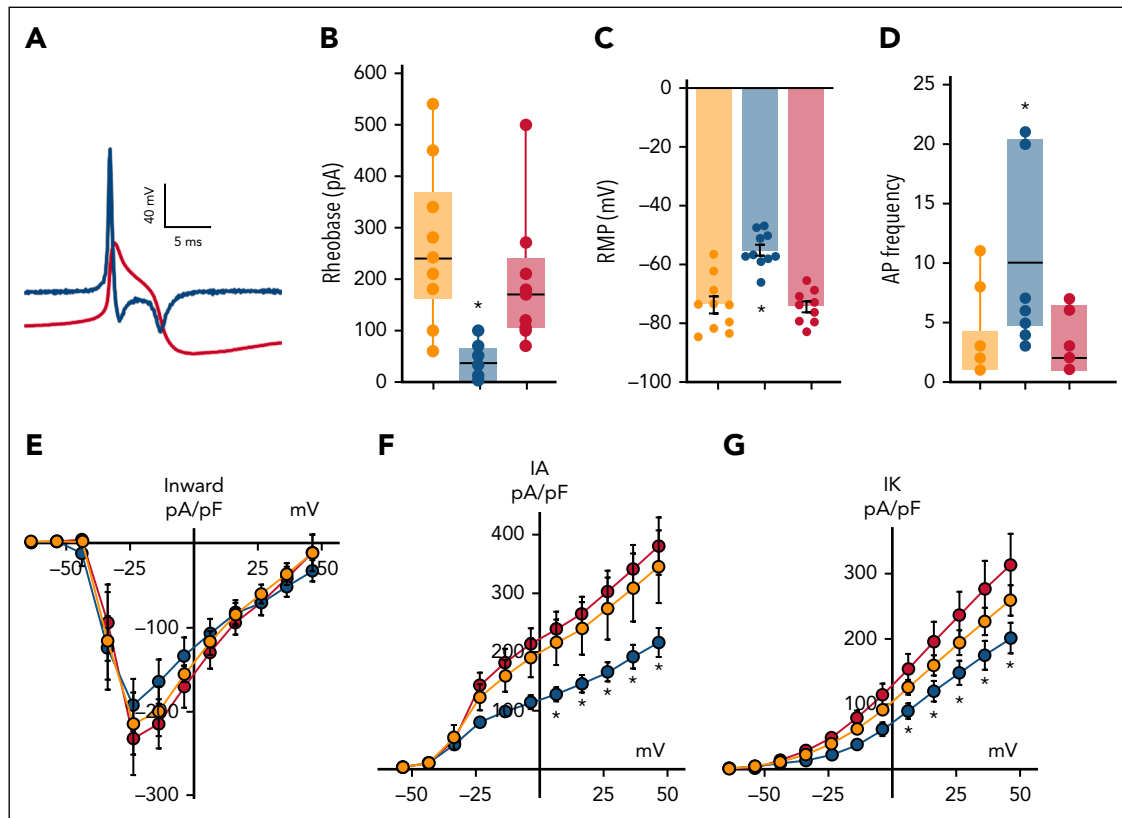


Figure 4. Sensitization of nociceptive DRG neurons in HbSS mice. Patch-clamp recording was performed on nociceptors acutely isolated from hyperalgesic HbSS or control (nonhyperalgesic) HbAA mice. (A) Nociceptors were identified by the presence of a shoulder in the repolarization phase of AP (red line), established using the first derivative of the AP waveform (blue line). (B) Nociceptors from hyperalgesic HbSS mice had lower rheobase (*different from other groups at $P = .006$, Kruskal-Wallis 1-way ANOVA on ranks with the Dunn test), (C) more positive RMP (*different from other groups at $P < .001$, 1-way ANOVA with Bonferroni test), and (D) more high frequency of AP (*different from other groups at $P < .05$, Kruskal-Wallis 1-way ANOVA on ranks with the Dunn test). All these parameters were normalized by treatment of HbSS mice with LPA₁R siRNA. (E) No changes were found in the values of inward currents in the nociceptors of HbSS mice compared with HbAA mice ($F[14,252] = 0.44$; 2-way RM ANOVA, $P = .96$; $n = 8$ -10 neurons per group). (F) Nociceptors from HbSS mice had reduced outward-inactivating ($F[42,462] = 2.83$; 2-way RM ANOVA, $P < .001$) and noninactivating currents (G) ($F[42,462] = 3.59$; 2-way RM ANOVA, $P < .001$). Functional knockdown of LPA₁R recovered both K⁺ currents. *Different from other groups at $P < .05$, Bonferroni test. For panels B-G, orange refers to HbAA mice, blue refers to hyperalgesic HbSS mice, and red refers to HbSS mice treated with LPA₁R siRNA.

effects of LPA on the central nervous system must be considered. Circulating LPA usually does not penetrate the blood-brain barrier but increases its permeability,⁷⁹ thereby enhancing the delivery of other pronociceptive mediators to the central nervous system that contribute to hyperalgesia in HbSS mice. Additional studies are needed to understand the central effects of circulating LPA on the development of chronic hyperalgesia in SCD and the possible contribution of LPARs located on neurons and glial cells.

Hyperalgesia in HbSS mice is a consequence of LPA₁R-mediated sensitization of DRG neurons, defined as a decrease in rheobase and an increase in RMP and AP frequency. The contribution of LPA₁R was confirmed by the restoration of these parameters after LPA₁R siRNA treatment. The combination of reduced rheobase with increased RMP strongly indicates LPA₁R-dependent inhibition of the 2-pore domain K⁺ channels, among which only TREK-1 (TWIK-related potassium channel-1) and TRAAK (TWIK-related arachidonic acid-stimulated K⁺ channel) are sensitive to LPA.⁴¹ TREK-1 and TRAAK channels in small DRG neurons are extensively colocalized with TRPV1 (transient receptor potential vanilloid 1),^{45,46} and membrane depolarization due to a decrease in K⁺ currents through these channels could increase responsiveness of

TRPV1-positive nociceptive neurons. The contribution of TRPV1 to pain was shown previously in both mice and patients with SCD.^{23,80}

Consistent with our previous report showing increased SA of C-fiber in HbSS mice in vivo,²⁴ 30% of the nociceptive DRG neurons isolated from hyperalgesic HbSS mice also exhibited spontaneous firing in vitro. This confirms the use of acutely isolated DRG neurons as a model reflecting their properties in vivo. Also, LPA-induced activation of sodium channels in DRG neurons has been demonstrated previously in vitro⁸¹; our studies did not show changes in inward currents in DRG neurons acutely isolated from hyperalgesic HbSS mice. This may be due to the ability of LPA to simultaneously enhance tetrodotoxin-resistant sodium currents and reduce tetrodotoxin-sensitive currents, which neutralizes the overall effect on total inward current⁸¹ and should be taken into account when considering sodium channel blockers for the treatment of chronic SCD pain. These results suggest that, despite the unaltered amplitude of AP in nociceptive neurons from HbSS mice due to stable inward currents, their sensitization is mediated by a decrease in outward K⁺ currents. Thus, targeting TREK-1 and TRAAK channels to prevent sensitization of nociceptors might represent a novel approach to reduce

chronic pain in SCD. Importantly, because RBCs do not express TREK-1 or TRAAK channels, their activation in neurons cannot affect the Gárdos channel (calcium-activated K⁺ channel) in RBCs and therefore cannot enhance their dehydration.⁸²

In conclusion, our results indicate that the ATX-LPA-LPA₁R signaling pathway is crucial for chronic hyperalgesia in SCD and may be a promising target for treatment. It is encouraging that ATX inhibitors (ClinicalTrials.gov identifier: NCT05483907, NCT05586516) and LPA₁R antagonists (NCT04308681 and NCT01651143) are actively undergoing clinical trials for fibrosis and associated inflammation. Moreover, the new, and highly potent and selective LPA₁R antagonist, ACT-1016-0707, with slow off-rate kinetics was recently reported to effectively suppress LPA₁R signaling even in the presence of high concentrations of LPA.⁸³ This represents an opportunity to rapidly assess the translational value of this study and determine whether blocking ATX-LPA-LPA₁R signaling is effective to reduce chronic pain in patients with SCD.

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Authorship

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lysophosphatidic acid (LPA) receptor expression, and writing of the manuscript; M.Y.G. and A.M.G. collected and analyzed data using ultraperformance liquid chromatography–tandem mass spectrometry to measure LPA in serum; M.J., L.N., K.T., and S.A. were involved in data collection for the behavioral studies; K.G. bred and provided the sickle mice and edited the manuscript; S.G.K. contributed to experimental design, data collection, and analyses for the electrophysiology studies in vivo, preparation of figures, and editing the manuscript; and D.A.S. was involved in experimental design, data collection for the electrophysiology studies in vivo, and writing of the manuscript.

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Footnotes

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